Supplementary Information: The Limits of Explanation

Drake Caraker, Bryan Arnold, David Rhoads

1 Detailed Domain Instances

Each of the eight domain instances derives the Rashomon property from first principles, requiring zero shared axioms. Full derivations are in the monograph; here we summarize the key construction for each domain.

1.1 Mathematics (Linear Algebra)

For an underdetermined system $Ax = b$ with $A \in \mathbb{R}^{m \times d}$, $m < d$, the null space $\ker(A)$ is $(d - m)$-dimensional. Any two solutions x_1, x_2 with $Ax_1 = Ax_2 = b$ but $x_1 \neq x_2$ witness the Rashomon property. The symmetry group is the continuous translation group $\mathbb{R}^{d-m}$ acting on the affine solution set. Empirically, pairwise RMSD between four standard solvers increases monotonically with null-space dimension ($p = 6.3 \times 10^{-9}$, Mann–Whitney U).

1.2 Biology (Codon Degeneracy)

Synonymous codons (e.g., UCU and UCC both encoding serine) witness the Rashomon property: same observable (amino acid), incompatible internal structure (nucleotide sequence). The symmetry group is S_k where k is the degeneracy level ($k \leq 6$). Across 120 eukaryotic cytochrome c sequences from NCBI RefSeq, codon usage entropy increases monotonically with degeneracy (Kruskal–Wallis $p = 3.5 \times 10^{-9}$); Met and Trp positions (degeneracy 1) have entropy identically zero.

1.3 Physics (Gauge Theory)

On a triangle graph with $\mathbb{Z}_2$ edge labels, gauge-equivalent configurations share the same holonomy (product of edge labels around cycles) but differ in local edge assignments. The symmetry group is $\mathbb{Z}_2^{|V|-1}$. Within-orbit variance of local edge assignments is nonzero, while holonomy (the gauge-invariant quantity) has zero variance within orbits.

1.4 Statistical Mechanics

Microstates (H, T) and (T, H) share the same macrostate (1 head) but differ in particle-level assignment. The symmetry group is S_Ω (permutations of particles). The microcanonical ensemble—uniform averaging over microstates consistent with a macrostate—is the G-invariant resolution.

1.5 Linguistics

The sentence "I saw the man with the telescope" admits two parse trees: $((V\ NP)\ PP)$ (instrument reading) and $(V\ (NP\ PP))$ (possessive reading). Both produce the same token sequence but commit to incompatible attachment structures. Across 50 ambiguous and 50 unambiguous

sentences parsed by four parsers (spaCy sm/md/lg, Stanza), pairwise UAS is significantly lower for ambiguous sentences ($p = 1.6 \times 10^{-3}$, Wilcoxon).

1.6 Crystallography

Phase retrieval maps signals to energy spectra via $|F|^2$, losing phase information. Signals $(1, 0)$ and $(0, 1)$ share the same energy spectrum but differ in phase. The ambiguity group $\mathbb{Z}_2 \times \mathbb{Z}_n \times \mathbb{Z}_2$ (conjugation, translation, reflection) acts on the signal space. Without positivity constraints, reconstruction RMSD diverges 1.5–1.8× more ($p = 3.4 \times 10^{-8}$, Mann–Whitney U).

1.7 Computer Science (Database Views)

A database view projects away hidden columns: rows (T, T) and (T, F) in the base table both project to (T) in the view. The symmetry group is S_k acting on hidden column values. Census disaggregation ambiguity (KL divergence from true county populations) scales with county count (Spearman $\rho = 0.70$, $p = 1.4 \times 10^{-8}$).

1.8 Statistics (Causal Discovery)

A causal chain $A \to B \to C$ and a fork $A \leftarrow B \to C$ encode the same conditional independences (Markov equivalence). The symmetry group consists of MEC-preserving edge orientation permutations. On the 8-node Asia network, causal discovery orientation agreement is only 0.33 at finite sample ($p = 4.4 \times 10^{-38}$, Mann–Whitney U).

2 Lean Formalization Details

The Lean 4 formalization (v4.30.0-rc1 with Mathlib) comprises 104 files containing 463 theorems and lemmas, 72 axioms, and 0 `sorry` (unproved goals). The proof complexity breaks down as: 71 core definitions and structures, 174 lemmas, 57 theorems of independent interest, and 12 tightness/witness constructions.

The axiom inventory is stratified by dependency: the core universal impossibility (`explanation_impossibility`) requires zero axioms beyond the Rashomon hypothesis. The 72 axioms serve domain-specific infrastructure: type declarations (Model, numTrees, attribution types), core GBDT properties (firstMover_surjective, proportionality_global), measure-theoretic infrastructure (modelMeasurableSpace, modelMeasure), and universal instance witnesses (AttentionConfig, etc.).

Key files: `ExplanationSystem.lean` (abstract impossibility), `AttributionInstance.lean` (SHAP/IG/LIME instance), `CausalInstance.lean` (DAG Markov equivalence), `MaximalIncompatibility.lean` (bilemma and tightness), `UniversalResolution.lean` (G-invariant resolution), `Necessity.lean` (biconditional: impossibility iff Rashomon). Full file listing and build instructions are in the repository `CLAUDE.md`.

3 Falsified Extensions

Three conjectured extensions were pre-registered and tested, all falsified:

Phase transition prediction. We conjectured a sharp phase transition in explanation stability at a critical Rashomon set size. Empirically, the transition is gradual (smooth sigmoid) rather than sharp, with no evidence of a critical threshold across five datasets and three model classes.

Tradeoff bound. We conjectured that the faithfulness–stability tradeoff follows a specific $1/k$ scaling law as the number of equivalent configurations k increases. The observed scaling is dataset-dependent and does not follow a universal power law.

Molecular evolution extrapolation. We conjectured that the character-theoretic prediction ($1/k$ information loss for S_k) would extend to predict absolute mutation rates. While the rank ordering of protein sequence diversity is predicted (Spearman $\rho = 0.88$), absolute rates require selection pressure parameters not captured by the symmetry framework.

These falsifications delineate the framework's proven domain (existence of instability, rank ordering, dose-response) from its speculative reach (quantitative scaling laws, absolute rates).

4 SAGE Algorithm

The Stability-Aware Grouped Explanation (SAGE) algorithm extends DASH to exploit the invariance counting law. Given P features in g correlation groups, SAGE: (1) identifies correlation groups via hierarchical clustering at threshold ρ_0; (2) computes group-level importance via orbit averaging within groups; (3) reports only the $\binom{g}{2}$ stably answerable between-group comparisons; (4) flags within-group comparisons as "structurally unreliable."

Audit results on 8 datasets (clinical, financial, genomics): SAGE reduces false discovery rate from 57% to <5% for feature ranking claims while preserving 100% of between-group ranking accuracy (in-sample $R^2 = 0.92$; leave-one-out CV $R^2 = 0.69$). The algorithm adds negligible computational overhead (<1% wall time) to any base attribution method.

5 CCA Spectrum and Continuous Symmetry

The character-theoretic formula $\eta = \dim(V^G)/\dim(V)$ quantifies the fraction of explanation space that is stably interpretable. For finite groups, $\dim(V^G)$ counts the multiplicity of the trivial representation in the decomposition of V; equivalently, it equals the number of orbits (by Burnside's lemma for permutation representations).

Across seven domain instances, the predicted η vs. observed instability rate yields $R^2 = 0.957$, slope $= 0.91$, intercept $= 0.008$ ($p = 1.4 \times 10^{-4}$). The spectral analysis reveals that the gap between the first and second singular values in the CCA decomposition corresponds to the G-invariant subspace dimension, providing a data-driven estimator of η when the group structure is unknown.

6 Noether Sensitivity Analysis

The invariance counting law predicts a bimodal flip-rate distribution: within-group pairs at ∼50% and between-group pairs at $< 2\%$. We verify this prediction is robust across the correlation range $\rho \in [0.50, 0.99]$:

ρ	Within-group flip	Between-group flip	Gap (pp)	p-value
0.50	0.49	0.018	47.2	< 0.0002
0.70	0.50	0.012	48.8	< 0.0002
0.90	0.50	0.005	49.5	< 0.0002
0.95	0.50	0.003	49.7	< 0.0002
0.99	0.50	0.001	49.9	< 0.0002

The gap is *invariant* across the entire correlation range—within-group flip rate remains at 50% regardless of correlation strength, while between-group flip rate decreases as correlation increases (features become more distinguishable at the group level). Cohen's d ranges from 1.41 ($\rho = 0.50$) to 2.89 ($\rho = 0.99$).

7 Multi-Model Validation

The Gaussian flip rate formula is validated across a 3×3 grid of model classes (XGBoost, RandomForest, Ridge) and datasets (California Housing, Adult Income, Diabetes). For each combination, 30 calibration models estimate (Δ, σ) per feature pair, and 30 independent validation models measure observed flip rates.

Out-of-sample R^2 ranges from 0.72 (Ridge on California Housing, where near-deterministic coefficients violate the Gaussian assumption) to 0.94 (XGBoost on Adult Income). The mean across the eight successful combinations is $R^2 = 0.848$. One combination (Ridge on California Housing) fails the Gaussianity check (Shapiro–Wilk $p < 0.001$) and is excluded; the formula's prediction conditional on Gaussianity holding is accurate.

8 Enrichment Tradeoff Details

The bilemma recovery via enrichment (adding a neutral element c to $\mathcal{H}$) introduces a faithfulness–informativeness tradeoff. On synthetic data with known ground truth: as the fraction of "tied" reports increases, stability improves monotonically but informativeness (fraction of decisive answers) decreases. The Pareto frontier is convex, with the G-invariant resolution sitting at the knee.

On real data (five clinical and financial datasets), DASH with varying tie thresholds traces the same convex frontier. At the default threshold, DASH achieves 95% stability with 38% decisiveness loss—the predicted tradeoff from the bilemma tightness analysis.

9 Gene Expression and High-Dimensional Results

On the AP_Breast_Lung cancer classification dataset ($P = 200$ genes selected by variance, $N = 253$ samples), 70% of gene importance comparisons are unreliable ($|\Delta|/\sigma < 0.5$) and only 4% are reliable ($|\Delta|/\sigma > 2$). SHAP values and gain-based importance show Spearman $\rho = 0.91$ for between-group rankings but $\rho = 0.12$ for within-group rankings, confirming the invariance counting prediction extends to high-dimensional biological data.

Across seven real datasets (2,062 feature-pair comparisons total), the aggregate statistics are: 62% unreliable, 34% intermediate, 4% reliable. The Gaussian flip rate formula achieves $R^2 = 0.83$ on gene expression data specifically, slightly below the overall mean of 0.848 due to heavier tails in the importance difference distribution.

10 Mechanistic Interpretability: Circuit Stability Experiment

10.1 Setup

Ten two-layer transformers (4 heads per layer, $d_{\text{model}} = 128$, MLP hidden dim = 512) are trained from *independent random initialisation* on modular addition ($a + b \bmod 113$), following Nanda et al. [2023]. Training: 50,000 steps, AdamW with lr = 10^{-3}, weight decay 1.0, full-batch (6,384

train / 6,385 test, 50/50 split). All 10 models reach 100% test accuracy (grokking). Circuit importance is measured by activation patching on 10 components (8 attention heads + 2 MLPs = $L_0H_0, \ldots, L_1H_3, \mathrm{MLP}_0, \mathrm{MLP}_1$) using 2,000 corrupted examples.

10.2 Primary Results

Pairwise Spearman ρ on 10-dimensional importance vectors across $\binom{10}{2} = 45$ model pairs:

Representation	Dim	Mean ρ	Min ρ	Perm. p
Full importance	10	0.518	-0.176	$< 10^{-6}$
G-invariant ($S_4 \times S_4$)	4	**0.929**	0.800	$< 10^{-6}$
Invariant excl. MLP1	3	0.822	0.500	$< 10^{-6}$
Heads only	8	0.131	-0.857	0.028

The symmetry group $G = S_4 \times S_4$ acts by within-layer head permutations (heads within each layer are architecturally identical). The G-invariant representation is $(\overline{\mathrm{imp}}_{L_0}, \overline{\mathrm{imp}}_{L_1}, \mathrm{imp}_{\mathrm{MLP}_0}, \mathrm{imp}_{\mathrm{MLP}_1})$ — mean head importance per layer plus MLP importances.

10.3 Per-Component Consensus

MLP1 is universally dominant (mean importance 9.60, CV = 0.027, rank std = 0.00 — always ranked #1). All 8 attention heads have CV = 0.32–0.66 and rank std = 1.2–2.6.

10.4 Noether Counting Within MI

Component pairs are classified by group structure: within-layer head pairs (12, S_4-symmetric), between-group pairs (33, involving different layers or MLPs).

Category	Pairs	Mean flip rate	Predicted
Within-layer (S_4)	12	**0.500**	0.500
Head vs MLP1	8	**0.000**	0.000
L_0 vs L_1 heads	16	0.421	$\sim$0.000
Head vs MLP0	8	0.094	$\sim$0.000

Within-layer flip rate = 0.500 exactly (std = 0.071, consistent with binomial noise at $\sqrt{0.25/45} = 0.075$). Mann-Whitney within > between: $p = 2.4 \times 10^{-5}$. The between-group prediction partially fails for L_0-vs-L_1 head pairs (0.421) because the two layers have overlapping importance ranges (mean $L_0 = 2.57$, mean $L_1 = 3.36$).

10.5 Controls

Corrected pre-grokking controls (matched $n_{\text{examples}} = 2000$):

Condition	Steps	Test acc	Mean ρ	p vs. post-grok
Random init	50	0.2%	0.300	— (baseline)
Memorised	500	$\sim$50%	0.668	—
Grokked	50,000	100%	0.518	2.0×10^{-4} (vs. random)

Cohen's d: post vs. random = 0.76 (medium-large); memorised vs. post = 0.58 (medium). Determinism check: same model patched twice yields $r = 0.9998$.

Fourier analysis. All 10 models learn completely different Fourier frequency representations (top-5 frequency Jaccard = 0.022). Each model uses a different subset of the ∼56 conjugate frequency pairs available in $\mathbb{Z}_{113}$, confirming genuine Rashomon: same function, different algorithms.

Perturbation dose-response. Adding Gaussian noise $\epsilon \cdot \|W\|_F$ per tensor: accuracy collapses between $\epsilon = 0.001$ (acc = 0.98, ρ vs. base = 0.995) and $\epsilon = 0.005$ (acc = 0.13). Inter-model L_2 distance is ∼40 vs. collapse perturbation L_2 of ∼13 (ratio ∼3×). Models occupy narrow basins in weight space.

10.6 η Law Reconciliation

$\eta = \dim(V^G)/\dim(V) = 4/10 = 0.40$. The η formula predicts instability $= 1 - \eta = 0.60$. Observed flip rate = 0.300. The discrepancy is reconciled by the Noether decomposition: overall flip rate $= (12/45) \times 0.500 + (33/45) \times 0.227 = 0.133 + 0.166 = 0.300$. The η formula counts dimensions lost; the flip rate counts pairwise ranking flips weighted by pair count — these are different metrics.

11 Gene Expression Pathway Divergence

The gene-expression alternation finding (Instance 1 in the main text) was tested across four microarray datasets (OpenML, Affymetrix HG-U133A platform):

Dataset	N	Top gene 1 (%)	Top gene 2 (%)	ρ	Mode	GO BP overlap
AP_Colon_Kidney	546	TSPAN8 (92%)	CEACAM5 (6%)	0.858	1	0 terms
AP_Endo_Colon	363	TSPAN8 (78%)	CEACAM5 (22%)	0.781	1	0 terms
AP_Breast_Colon	1,545	COL3A1 (72%)	COL1A1 (28%)	0.843	2	3 terms
AP_Breast_Lung	470	SFTPB (98%)	SFTPA1 (2%)	—	ctrl	control

Mode 1 (distinct pathways). TSPAN8 (tetraspanin 8): integrin binding, cell motility, invasion/metastasis. CEACAM5 (carcinoembryonic antigen): cell–cell adhesion, immune signalling, apoptosis regulation, classical clinical colon cancer marker. Gene Ontology Biological Process overlap: 0 terms (TSPAN8 has 1 BP term; CEACAM5 has 6). The two genes serve distinct cancer biology roles; which pathway a drug-discovery pipeline targets depends on the training seed.

Mode 2 (same pathway, redundant markers). COL3A1 and COL1A1 are both type I/III collagens in the extracellular matrix pathway, sharing 3 GO BP terms (animal organ development, collagen fibril organization, system development). Their alternation reflects classical collinearity instability within a single pathway.

Controls. Under deterministic training (subsample = 1.0), all datasets produce a single ranking with 100% agreement. Model test accuracies on AP_Colon_Kidney: mean 96.4%, std 0.5%, range [95.1%, 97.6%] ($n = 50$) — confirming near-identical predictive performance across seeds. The DASH ensemble average places both TSPAN8 and CEACAM5 in the consensus top-2.

12 Neuroimaging Multi-Analyst Reanalysis

The NARPS reanalysis (Instance 4) downloaded 48 of 70 teams' unthresholded t-statistic maps from NeuroVault and parcellated with the Schaefer 100-parcel atlas (7 Yeo resting-state networks). Key methodological details:

Activation-level control. The raw Noether effect ($d = 0.43$) was tested against the activation-level confound: within-network regions have similar activation levels, potentially explaining similar disagreement. Regressing $|\Delta\text{overlap}|$ from disagreement similarity reduces d to 0.32 ($p = 8.1\times10^{-19}$, bootstrap CI [0.006, 0.008]). Activation similarity explains $R^2 = 0.67$ of overall disagreement variance; the network effect persists as a 26% reduction in d.

Controls. Random grouping null ($d = 0.001 \pm 0.050$, permutation $p < 0.001$); spatial autocorrelation (k-means $d = 0.07$–0.19 across $k = 5, 7, 10, 14$, all below network $d = 0.43$); per-hypothesis consistency (7/7 non-empty hypotheses show the effect); Schaefer-400 replication ($d = 0.41$).

Convergence. Split-half Pearson correlation exceeds 0.95 at $M = 16$ [bootstrap 95% CI: 10–22]. Convergence rate: $1 - a/M^{1.0}$ ($R^2 = 0.97$).

Aggregation method comparison (all at $M = 48$). Mean: faithfulness 0.669, unfaithfulness 0.739. Median: 0.660, 0.719. Trimmed mean: 0.668, 0.727. Weighted mean: 0.664, 0.725. All within 3%.

13 Classification of Impossibility Theorems

Extended Data Table 2 classifies 20 impossibility theorems by tightness type. Full details: 6 instances have structural Lean proofs (non-trivial solution spaces); 14 use trilemma models (3-element types). The classification, tightness type exhaustiveness/exclusivity proofs, enlargement monotonicity, and obstruction taxonomy are formalised in the `UnifiedMetaTheorem.lean` file (companion Ostrowski repository: 240 theorems, 10 axioms, 0 sorry).

Bell (CHSH) and Kochen-Specker are naturally 2-ary impossibilities; their 3-property decomposition into locality/determinism/correctness (Bell) and row-assignment/column-assignment/consistency (KS) is decomposition-dependent. Alternative decompositions may yield different tightness types.

The quantum linearity trilemma (universality + amplification + informativeness) is formalised as a 3-element model, not a structural proof over all quantum channels. Structural formalization would require formalizing CPTP maps in Lean — a substantial future project.

14 Approximate Rashomon Extension

The approximate Rashomon extension relaxes exact observational equivalence to ϵ-approximate equivalence: $d(obs(\theta_1), obs(\theta_2)) < \epsilon$. The Lean formalization (`ApproximateEquity.lean`) proves that bounded proportionality (a weakening of exact proportionality) implies the Rashomon property, bridging the gap between the theoretical exact-equivalence setting and practical approximate-equivalence scenarios.

The key result is that for any $\epsilon > 0$, if the system has ϵ-approximate Rashomon witnesses, the impossibility holds with a degradation factor bounded by $O(\epsilon)$ in the faithfulness guarantee.

This ensures the theorem's practical relevance: real-world systems with near-identical (not exactly identical) predictions still face the fundamental tradeoff.

References

Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. *arXiv preprint arXiv:2301.05217*, 2023.